

PROJECT

5.7

VPC MONITORING WITH FLOW LOGS

NETWORKING SERIES

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Objective

This project focuses on setting up two VPCs and testing a peering connection between them. It enables private communication across VPCs without using the internet.

Next, it sets up VPC Flow Logs to capture traffic data and uses CloudWatch Log Insights to analyze patterns, helping monitor and secure the network efficiently.

Services

Amazon VPC - Creates isolated networks where you launch AWS resources.

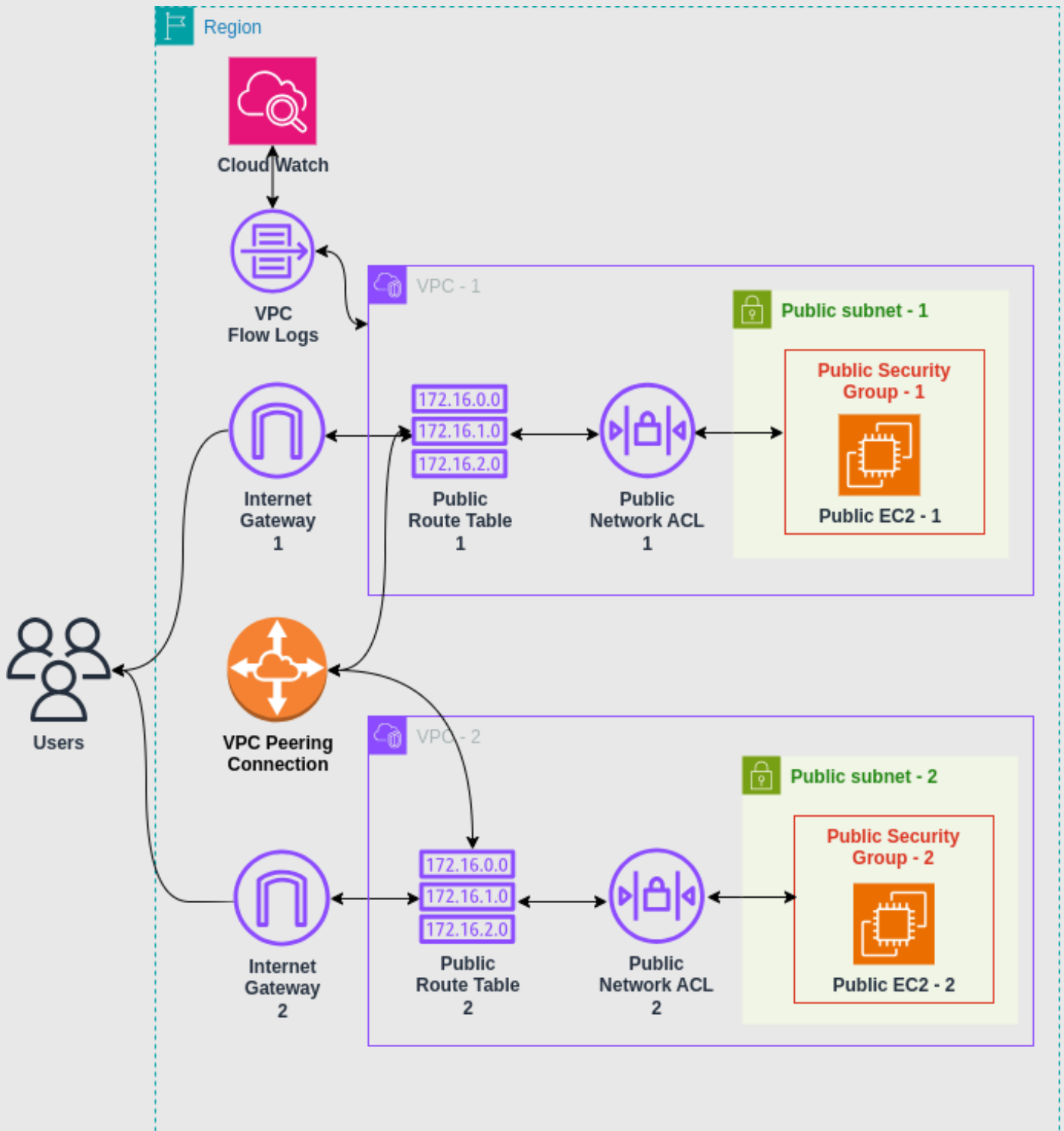
VPC Peering - Connects two VPCs to enable private traffic flow between them.

VPC Flow Logs - Captures detailed information about IP traffic within your VPC.

Amazon CloudWatch - Monitors and visualizes logs and metrics from AWS resources.

CloudWatch Log Insights - Allows you to query and analyze VPC Flow Logs for network insights.

Architecture Diagram



The architecture includes two VPCs connected through a VPC Peering link for private communication. Each VPC has subnets and instances generating traffic. VPC Flow Logs are enabled to capture network data, which is sent to CloudWatch Logs. Using CloudWatch Log Insights, traffic patterns and security events are analyzed efficiently.

Steps to Implement

1. Create Two VPCs:

Start by creating two separate VPCs with non overlapping CIDR blocks (like 10.0.0.0/16 and 20.0.0.0/16). These VPCs represent two isolated cloud networks. This gives you control over separate environments that can be securely connected later.

2. Set Up VPC Peering:

Create a VPC peering connection between the two VPCs to enable communication. Accept the request from the second VPC and update route tables on both sides. This lets instances in one VPC talk to those in the other privately.

3. Enable VPC Flow Logs:

Enable Flow Logs for each VPC to capture network traffic data. Choose CloudWatch Logs as the destination to view and analyze logs. This helps you monitor what's happening inside your network.

4. Analyze with CloudWatch Log Insights:

Open CloudWatch Log Insights and select the log group for each VPC. Run queries to examine traffic patterns or detect suspicious behavior. This step ensures transparency, security, and easier troubleshooting.

Story Time

VPCs (Two Cloud Cities):

Build two separate cloud kingdoms VPC 1 and VPC 2 each with their own city walls and neighborhoods. These cities exist side by side but can't visit each other yet. Time to build a secret bridge!

VPC Peering (The Secret Bridge):

Create a magical secret bridge that connects the two cloud cities. Once the bridge is accepted and maps are updated, citizens can travel safely between kingdoms. Now the cities can share resources without any gatekeepers!

VPC Flow Logs (The City Watch Scrolls):

Set up the City Watch to keep magical scrolls recording all traffic coming in and out of each city. These scrolls are sent to the wise wizard, CloudWatch, for safekeeping. This helps spot any strange visitors or unusual events.

CloudWatch Log Insights (The Wizard's Spyglass):

Use the wizard's spyglass to read the City Watch scrolls and spot suspicious activity or traffic patterns. With clever spells (queries), the wizard helps you keep both cities safe and running smoothly.

BY
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THANK YOU!

